

• PREMIUM CAREER ROADMAP

SDET Interview Master Curriculum

2.5 – 3 Years Experience

A complete, navigable syllabus of every topic an SDET must master — structured module by module, the way a Principal SDET mentor would map your entire preparation.

Core Java & Java 8

Problem Solving

Selenium & Playwright

API Automation

SQL

DevOps & CI/CD

System Design

Behavioral

Mock Interviews

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9 Modules • 6 Trackers • One Roadmap

The Curriculum

01	Programming Fundamentals Core Java & Java 8	40 topics 40–50 hrs
02	Problem Solving Data Structures, Algorithms & Coding	19 topics 50–60 hrs
03	UI Automation Selenium · Playwright · Framework Design	55 topics 45–55 hrs
04	API Automation REST · REST Assured · Postman	38 topics 30–40 hrs
05	Database SQL for SDETs	24 topics 20–25 hrs
06	DevOps & CI/CD Pipelines, Containers & Cloud	15 topics 20–30 hrs
07	System Design Test Strategy & Architecture	15 topics 25–30 hrs
08	Behavioral Interviews Story, Impact & Negotiation	9 topics 10–15 hrs
09	Mock Interview Preparation Simulate · Rehearse · Refine	8 topics 15–20 hrs
★	Appendix · Trackers & Checklists Skills · Completion · Progress · Readiness · Learning Order · Revision	6 tools

9 MODULES
223 TOPICS
6 TRACKERS

Open this book, pick a topic, and know exactly what to cover.

INTERVIEW WEIGHTAGE

Very High

25–30%

ESTIMATED STUDY TIME

40–50 hrs

focused effort

TOTAL TOPICS

40

to master

OBJECTIVE

Build unshakeable command over Core Java and Java 8 so you can write clean, efficient, production-grade automation code.

WHY INTERVIEWERS ASK THIS

Java is the backbone of most SDET frameworks. Interviewers use it to judge coding maturity, OOP thinking, and whether you understand what runs under the hood.

PREREQUISITES

Basic programming logic

LEARNING OUTCOME

Explain JVM internals, apply OOP and Java 8 features fluently, and solve language-level problems on the spot.

Complete Topic List

◆ **Java Basics** 12

JDK

JRE

JVM

Java Architecture

Compilation Process

Variables

Data Types

Operators

Control Statements

Methods

Arrays

Strings

◆ **Object-Oriented Programming** 8

Constructors

this

super

Inheritance

Polymorphism

Abstraction

Interfaces

Encapsulation

◆ **Core Java APIs** 4

Exception Handling

Collections Framework

Generics

File Handling

◆ **Java 8** 5

Lambda Expressions

Functional Interfaces

Streams API

Optional

Method References

◆ **Multithreading & Concurrency** 5

Multithreading

Synchronization

Executor Framework

Callable

Future

◆ **JVM & Performance** 4

JVM Memory

Garbage Collection

Memory Management

Performance Optimization

Interview Topics

Mini Assignments

INTERVIEW WEIGHTAGE

High

15–20%

ESTIMATED STUDY TIME

50–60 hrs

focused effort

TOTAL TOPICS

19

to master

OBJECTIVE

Develop the DSA and logical reasoning needed to clear live coding rounds at product companies.

WHY INTERVIEWERS ASK THIS

Interviewers test raw problem-solving, edge-case handling, and how you think under time pressure — not just framework recall.

PREREQUISITES

Module 1 (Core Java, Collections)

LEARNING OUTCOME

Solve easy–medium problems across strings, arrays, and core data structures within interview time limits.

Complete Topic List

◆ Fundamentals 4

Numbers

Patterns

Strings

Arrays

◆ Data Structures 5

HashMap

Stack

Queue

Linked List

Trees

◆ Algorithms 7

Searching

Sorting

Binary Search

Sliding Window

Two Pointer

Recursion

Backtracking Basics

◆ Applied Coding 3

Java 8 Coding

Collection Coding

Interview Coding

MODULE 03

03

UI Automation

Selenium · Playwright · Framework Design

INTERVIEW WEIGHTAGE

Very High

20–25%

ESTIMATED STUDY TIME

45–55 hrs

focused effort

TOTAL TOPICS

55

to master

OBJECTIVE

Master Selenium and Playwright, then design a scalable automation framework from scratch.

WHY INTERVIEWERS ASK THIS

UI automation is the daily job of an SDET. Interviewers probe depth — waits, locators, framework design — to separate scripters from engineers.

PREREQUISITES

Module 1

LEARNING OUTCOME

Automate complex web flows, debug flaky tests, and architect a maintainable, reusable framework.

Complete Topic List

SECTION A Selenium

18 topics

Architecture WebDriver Locators XPath CSS Selectors Waits Frames Alerts

Windows Actions JavaScript Executor Shadow DOM Relative Locators Grid

Parallel Execution Best Practices Interview Questions Mini Project

SECTION B Playwright (JavaScript)

21 topics

Architecture Project Structure Locators Assertions Fixtures Hooks Authentication

Storage State API Testing Network Interception Network Mocking File Upload File Download

Parallel Execution Reports Trace Viewer Debugging CI/CD Integration Best Practices

Interview Questions Mini Project

SECTION C Automation Framework Design

16 topics

Framework Architecture POM Hybrid Framework Keyword Driven Data Driven Factory Pattern

Singleton Pattern Retry Logic Logger Reporting Utilities Configuration Management

Test Data Management

Parallel Execution

Folder Structure

Coding Standards

INTERVIEW WEIGHTAGE

High

15–20%

ESTIMATED STUDY TIME

30–40 hrs

focused effort

TOTAL TOPICS

38

to master

OBJECTIVE

Test and automate REST, SOAP, and GraphQL APIs using Postman and REST Assured with strong fundamentals.

WHY INTERVIEWERS ASK THIS

Modern testing shifts left to the API layer. Interviewers check whether you truly understand HTTP, auth, and contract-level validation.

PREREQUISITES

Module 1 · HTTP basics

LEARNING OUTCOME

Design end-to-end API test suites with auth, chaining, schema validation, and CI integration.

Complete Topic List

◆ API Fundamentals

12

REST

SOAP

GraphQL Basics

HTTP

HTTPS

Methods

Status Codes

Headers

Request

Response

Cookies

Sessions

◆ Authentication

4

Basic Authentication

Bearer Token

OAuth 2.0

JWT

◆ Data Handling

7

Serialization

Deserialization

JSON

XML

JSON Schema Validation

API Chaining

CRUD Operations

◆ Tools & Assertions

7

Postman

REST Assured

Request Specification

Response Specification

Logging

Assertions

Error Handling

◆ Advanced Concepts

6

Rate Limiting

Pagination

Versioning

Contract Testing

Mock APIs

Performance Basics

Interview Questions

Mini Project

INTERVIEW WEIGHTAGE

Medium-High

10-15%

ESTIMATED STUDY TIME

20-25 hrs

focused effort

TOTAL TOPICS

24

to master

OBJECTIVE

Write and optimize SQL for backend validation and data-driven testing.

WHY INTERVIEWERS ASK THIS

SDETs validate data integrity constantly. Joins, subqueries, and query logic appear in almost every interview round.

PREREQUISITES

None

LEARNING OUTCOME

Write complex queries — joins, CTEs, window functions — and reason about performance and transactions.

Complete Topic List

◆ SQL Fundamentals

4

DDL

DML

DCL

TCL

◆ Query Building

8

SELECT

WHERE

GROUP BY

HAVING

ORDER BY

CASE

Functions

Aggregate Functions

◆ Advanced SQL

7

Joins

Subqueries

Views

Indexes

CTE

Window Functions

Stored Procedures

◆ Design & Performance

3

Normalization

Transactions

Performance Optimization

◆ Practice

2

Interview Questions

Hands-on Queries

INTERVIEW WEIGHTAGE

Medium

8-12%

ESTIMATED STUDY TIME

20-30 hrs

focused effort

TOTAL TOPICS

15

to master

OBJECTIVE

Integrate automated tests into modern delivery pipelines using Git, CI tools, and containers.

WHY INTERVIEWERS ASK THIS

Automation only matters if it runs continuously. Interviewers assess whether you can operationalize your suites.

PREREQUISITES

Module 3 or 4

LEARNING OUTCOME

Configure pipelines, containerize tests, and manage branching strategies confidently.

Complete Topic List

◆ Version Control

7

Git

GitHub

Git Flow

Branching Strategy

Merge

Rebase

Cherry Pick

◆ CI/CD & Build

3

GitHub Actions

Jenkins

Maven

◆ Containers & Orchestration

3

Docker Basics

Docker Compose

Kubernetes Basics

◆ Cloud & OS

2

AWS Basics

Linux Basics

INTERVIEW WEIGHTAGE

Medium-High

10-15%

ESTIMATED STUDY TIME

25-30 hrs

focused effort

TOTAL TOPICS

15

to master

OBJECTIVE

Think like an architect about test strategy, framework design, and scalable quality systems.

WHY INTERVIEWERS ASK THIS

At 2.5-3 years, interviewers expect design thinking — test pyramids, patterns, SOLID — not just execution.

PREREQUISITES

Modules 1, 3, 4

LEARNING OUTCOME

Present a coherent automation architecture and defend design decisions in senior rounds.

Complete Topic List

◆ Test Strategy 5

Automation Strategy

Automation Architecture

Test Pyramid

Testing Strategy

Microservices Testing

◆ Design Patterns 5

Factory

Singleton

Strategy

Builder

Observer

◆ Principles & Design 3

SOLID Principles

Low Level Design

High Level Design

◆ Quality Attributes 2

Scalability

Maintainability

Behavioral Interviews

Story, Impact & Negotiation

INTERVIEW WEIGHTAGE

Medium

8-12%

ESTIMATED STUDY TIME

10-15 hrs

focused effort

TOTAL TOPICS

9

to master

OBJECTIVE

Communicate impact, ownership, and collaboration through structured, confident storytelling.

WHY INTERVIEWERS ASK THIS

Technical skill gets you shortlisted; behavioral answers decide the offer and the level.

PREREQUISITES

Polished resume · 2-3 strong project stories

LEARNING OUTCOME

Deliver crisp STAR-format answers, handle HR rounds, and negotiate offers effectively.

Complete Topic List

◆ Personal Branding 2

Resume

LinkedIn

◆ Storytelling & Impact 5

STAR Method

Project Explanation

Leadership

Conflict Resolution

Ownership

◆ HR & Negotiation 2

Salary Negotiation

HR Questions

Mock Interview Preparation

Simulate · Rehearse · Refine

INTERVIEW WEIGHTAGE

Applied

Final Phase

ESTIMATED STUDY TIME

15-20 hrs

focused effort

TOTAL TOPICS

8

to master

OBJECTIVE

Simulate real interview conditions across every round to build speed, composure, and consistency.

WHY INTERVIEWERS ASK THIS

Preparation without pressure-testing fails on interview day. Mocks expose gaps early, while there is still time to fix them.

PREREQUISITES

Modules 1-8

LEARNING OUTCOME

Walk into interviews calm, rehearsed, and company-ready.

Complete Topic List

◆ Full Mock Rounds

8

Java

Coding

Automation

API

SQL

System Design

Behavioral

Company Specific



APPENDIX

Trackers & Checklists

Your progress command centre

Master Skills Checklist

- | | |
|--|--|
| ☐ Core Java + OOP fluency | ☐ Design patterns in automation |
| ☐ Java 8 (Streams, Lambdas, Functional Interfaces) | ☐ API testing (REST Assured / Postman) |
| ☐ Collections & Generics | ☐ Auth flows (OAuth2, JWT, tokens) |
| ☐ Multithreading & Concurrency | ☐ JSON schema & contract testing |
| ☐ JVM internals & GC | ☐ SQL joins, subqueries, CTEs, window functions |
| ☐ Problem solving (arrays, strings, maps) | ☐ Git & branching strategy |
| ☐ Core data structures (stack, queue, linked list, trees) | ☐ CI/CD pipelines (Jenkins / GitHub Actions) |
| ☐ Searching, sorting & binary search | ☐ Docker & basic Kubernetes |
| ☐ Sliding window & two pointer | ☐ Test strategy & test pyramid |
| ☐ Selenium (locators, waits, actions) | ☐ SOLID principles & LLD/HLD |
| ☐ Playwright end-to-end | ☐ STAR-format behavioral answers |
| ☐ Framework design (POM, data-driven, hybrid) | ☐ Salary negotiation |

Topic Completion Tracker

Module	Total Topics	Completed	%
M01 • Programming Fundamentals	40		
M02 • Problem Solving	19		
M03 • UI Automation	55		
M04 • API Automation	38		
M05 • Database	24		
M06 • DevOps & CI/CD	15		
M07 • System Design	15		
M08 • Behavioral Interviews	9		
M09 • Mock Interview Preparation	8		

Study Progress Tracker

Module	Est. Time	Status	Notes
M01 • Programming Fundamentals	40–50 hrs	Not Started	
M02 • Problem Solving	50–60 hrs	Not Started	
M03 • UI Automation	45–55 hrs	Not Started	
M04 • API Automation	30–40 hrs	Not Started	
M05 • Database	20–25 hrs	Not Started	
M06 • DevOps & CI/CD	20–30 hrs	Not Started	
M07 • System Design	25–30 hrs	Not Started	
M08 • Behavioral Interviews	10–15 hrs	Not Started	
M09 • Mock Interview Preparation	15–20 hrs	Not Started	

Interview Readiness Tracker

Rate your confidence per round from 1 to 5 stars. Only book the real interview when every row is 4★ or higher.

Round	Confidence (1-5)	Gaps to close
Core Java	☆ ☆ ☆ ☆ ☆	
Coding / DSA	☆ ☆ ☆ ☆ ☆	
UI Automation	☆ ☆ ☆ ☆ ☆	
API Automation	☆ ☆ ☆ ☆ ☆	
SQL / Database	☆ ☆ ☆ ☆ ☆	
System Design	☆ ☆ ☆ ☆ ☆	
Behavioral / HR	☆ ☆ ☆ ☆ ☆	

Recommended Learning Order

#	Module	Focus	Why this order
1	Module 1	Programming Fundamentals	Foundation for everything else
2	Module 2	Problem Solving	Runs in parallel with Module 1
3	Module 5	Database	Low dependency — build early
4	Module 4	API Automation	Shift-left layer, high ROI
5	Module 3	UI Automation	Selenium → Playwright → Framework
6	Module 6	DevOps & CI/CD	Operationalize your suites
7	Module 7	System Design	Requires 1, 3 & 4 first
8	Module 8	Behavioral Interviews	Prepare stories throughout
9	Module 9	Mock Interviews	Final pressure-test phase

Final Revision Checklist

The last 48 hours. Do these, then rest.

- ☐ Re-solve 20 core coding problems from memory
- ☐ Revise Java 8 Streams & Collections one-liners
- ☐ Walk through your framework architecture out loud
- ☐ Rehearse 5 STAR stories (2 min each)
- ☐ Revise HTTP status codes, auth flows & REST Assured syntax
- ☐ Practice 10 SQL queries (joins + window functions)
- ☐ Sketch the test pyramid & one automation architecture
- ☐ Review Git rebase vs merge & branching strategy
- ☐ Re-read the target company's tech stack & JD
- ☐ Prepare 3 questions to ask the interviewer
- ☐ Confirm resume claims match your stories
- ☐ Sleep well — composure beats cramming

Master the map. Then walk it with confidence.

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